

The MediaTek logo, consisting of the word "MEDIATEK" in a bold, sans-serif font, is centered within a white, parallelogram-shaped background.

DRAM Components Policy

IoT Team

Version History

Version	Date	Author (Optional)	Description
1.4	2023-03-24		Official release
1.5	2025-03-19		Add new page 13

Agenda

- ❑ DRAM Components Policy
- ❑ New DRAM Component (C**V**I Flow)
- ❑ New DRAM Component (C**Q**I Flow)

DRAM Components Policy

*QVL: Qualified Vendor List

*CVI: Customer **Validation** Item

*CQI: Customer **Qualified** Item

Purpose	Direct Customer	Indirect Customer
1. New Project KO	*Must follow MMD and select the components in the QVL system. (No need to apply for CVI or CQI flow)	
2. PCB Layout Quality	*Must: pass PCB Layout Quality - Power Delivery Network Simulation - Compare the DRAM Trace Layout fully same with MMD	
3. PCBA Bring up	Without Problems	
4. PCB Layout is changed Our DRAM chip is changed	CVI Flow	CQI Flow
	- MTK provides materials (Document, Tool) - Prepares chamber - Prepares more than 4 boards with thermal pad	- MTK provides materials (Document, Tool) - Prepares chamber - Prepares more than 4 boards with thermal pad
	- Direct Customer Q&A by eService - Support All Platforms - Pass result will list in MediaTek QVL system	- Indirect Customer Q&A by VAR support - Support Platform: MT8365, MT8390, MT8395 - Pass result will not list in MediaTek QVL system - Additional Document: JEDEC Judgement method

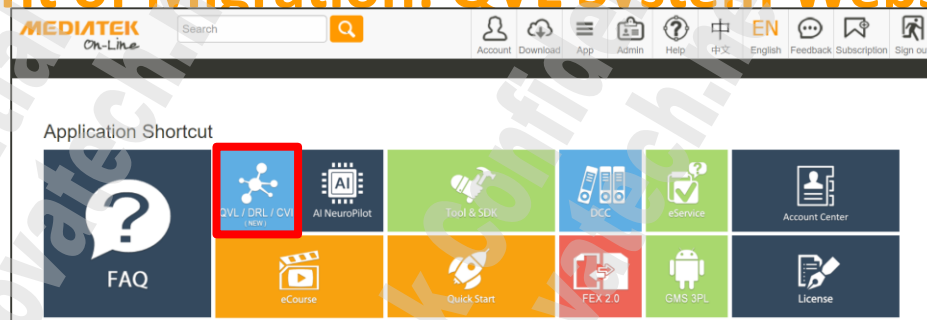
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QVL System

should be

- ❑ MOL Portal provides a qualified vendor list to speed up product design.

Announcement of Migration: QVL System Website



Mediatek On-Line

Home

New QVL Request

Workspace

Maintain QVL

Component Management

Search Zone

Product Line :
IoT

Discrete LPDDR4

Discrete LPDDR4X

PCDDR4

Change Mark	Attachments	Vendor	Vendor P/N	Package	RAM Type	RAM Size	DRAM Description	Standby Current(mA)	Carried Out Type	Chip	Quality Status
		Samsung	K4F6E3S4HM-MGCJ	200_FBGA	Discrete LPDDR4	16Gb	x32, VDDQ=1.1V		QVL	MT8365	Qualified
		Samsung	K4FBE3D4HM-MGCJ	200_FBGA	LPDDR4	32Gb	x32, VDDQ=1.1V		CVI	MT8365	OnGoing
		Micron	MT53E512M3ZD1NP-046 WTR	200_FPGA	LPDDR4	2GB	x32, VDDQ=1.1V		CVI	MT8365	Qualified

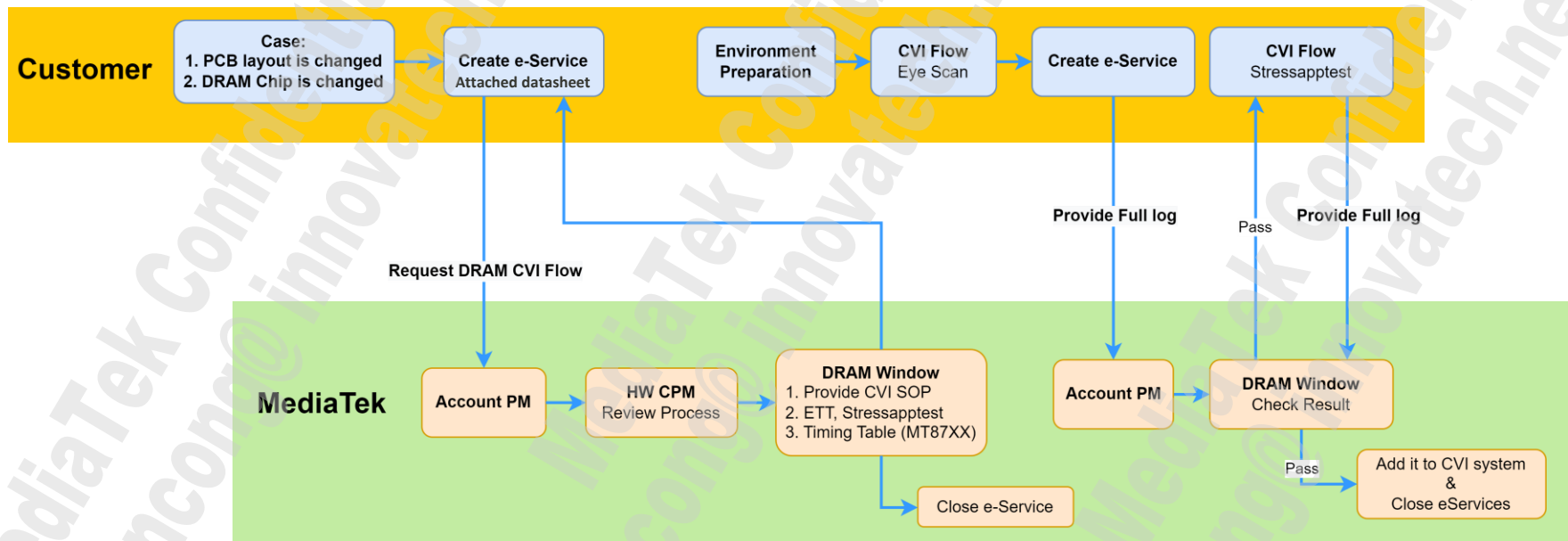
MT8365

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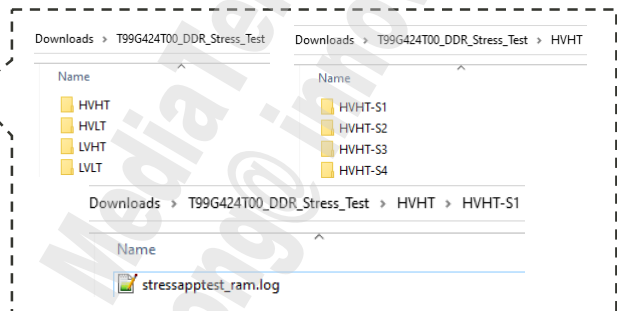
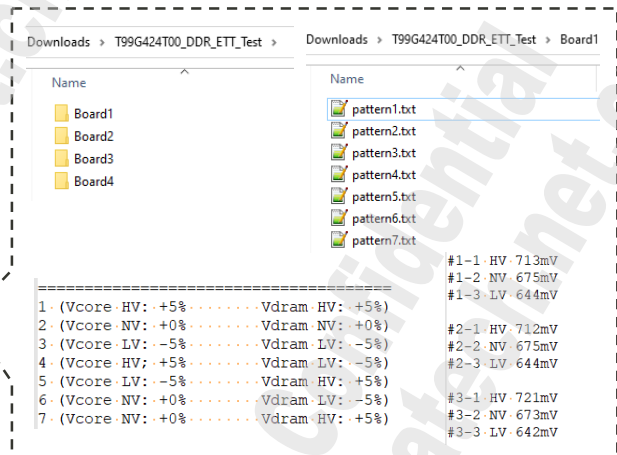
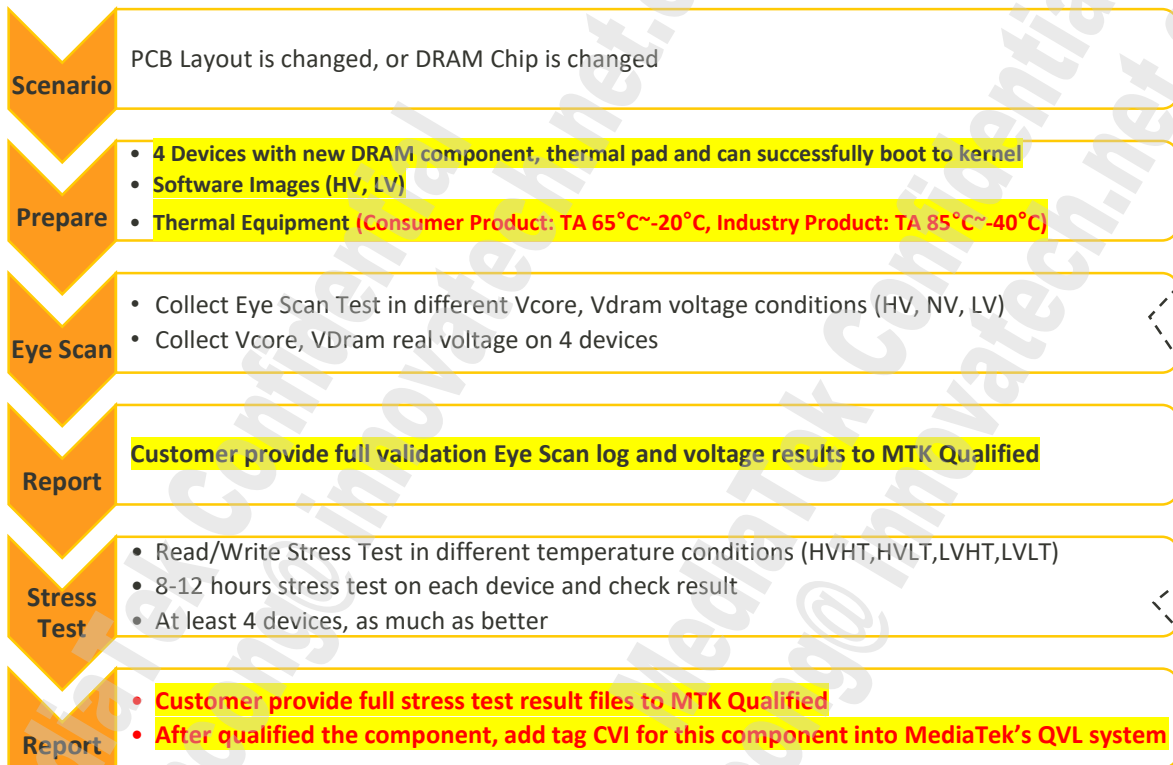
New DRAM Component: CVI Flow

Co-working Model (DRAM CVI Flow)

- ***Note: If the PCB Layout doesn't follow MMD, and Layout Quality failed. (No Support on CVI Flow)**
- **Support All Platforms**



DRAM CVI Flow

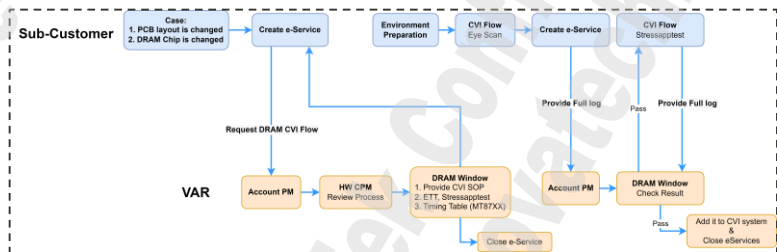
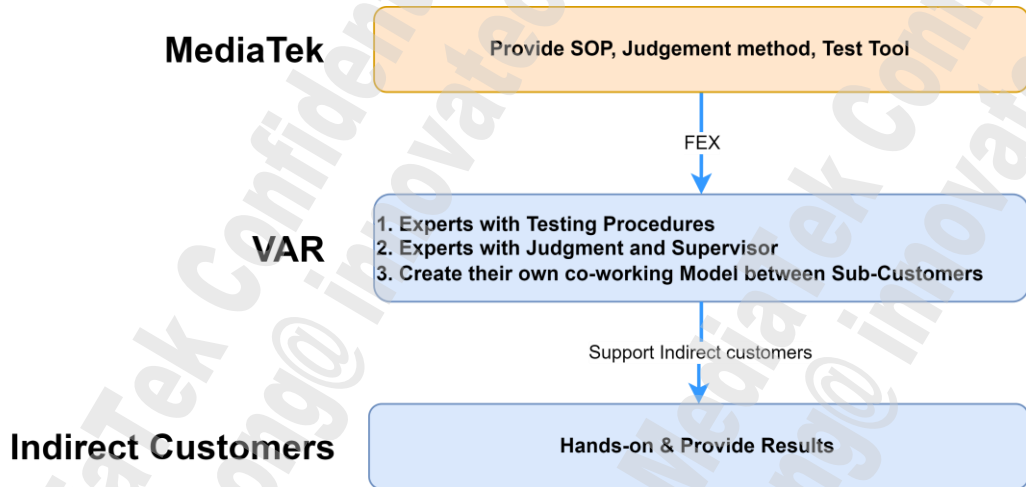


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New DRAM Component: CQI Flow

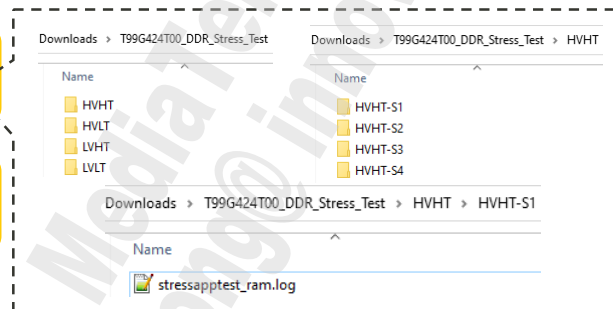
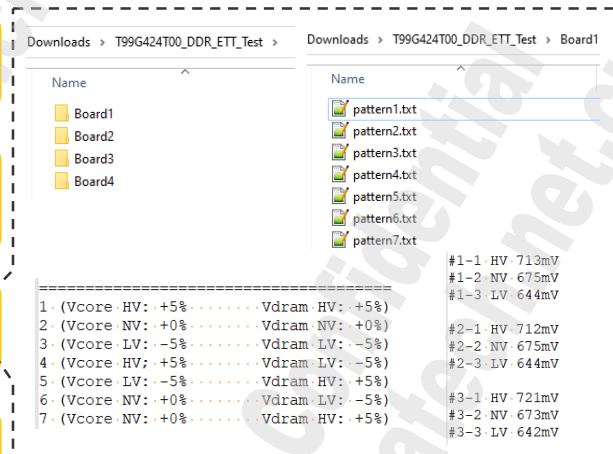
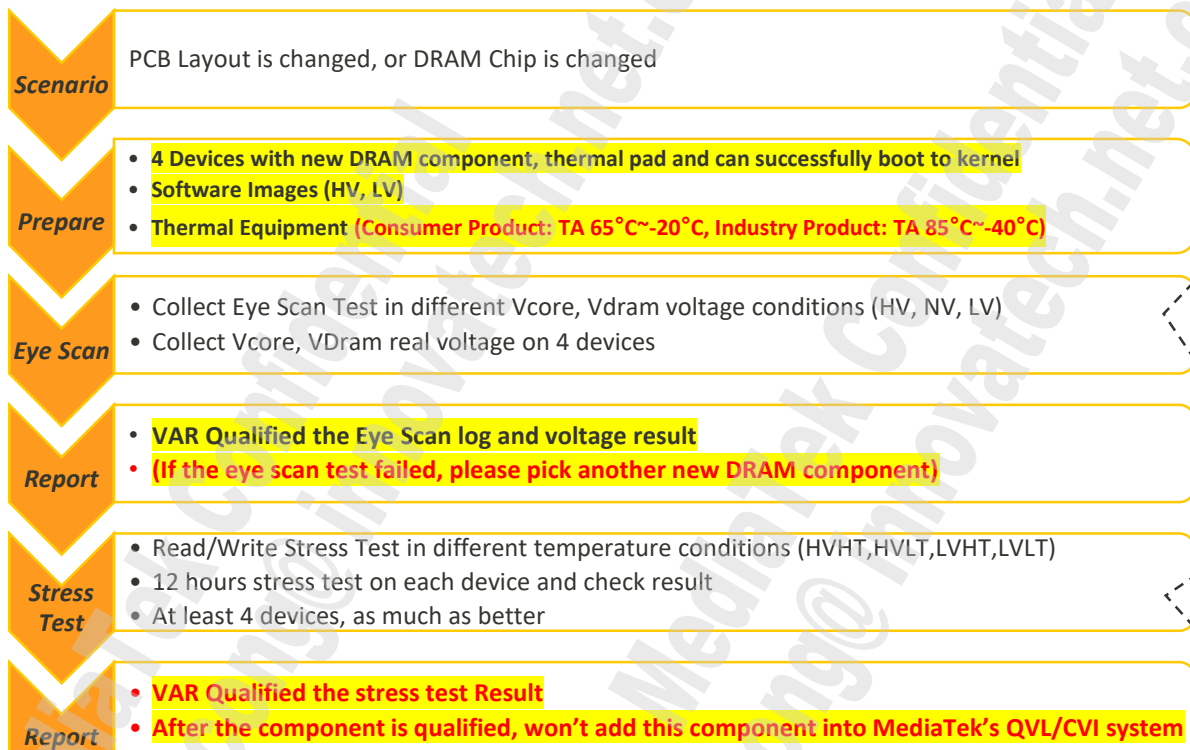
Co-working Model (DRAM CQI Flow)

- *Note: If the PCB layout doesn't follow MMD, and the layout qualification fails, there will be no support on CQI Flow.**
- Support Platforms: MT8365, MT8390, MT8395**



Concept Reference

DRAM CQI Flow



DRAM CVI & CQI Documents & Tool

- The modem has no plan to provide it.
- The Genio series can be downloaded from MOL Tools (IoT Product Line)

MEDIATEK Online Tool

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Tool Name:

Product Line: IoT

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Name	Version	Release Date	Product Line	Download
☐ Genio I200 MT8395 Yocto DRAM CVI CQI Tools	V1.2	2025-03-18	IoT	PDF JLS EXE SRC
☐ Genio I200 MT8395 Android DRAM CVI CQI Tools	V1.3	2025-03-18	IoT	PDF JLS EXE SRC
☐ Genio 700 510 MT8390 MT8370 Yocto DRAM CVI CQI Tools	V1.4	2025-03-18	IoT	PDF JLS EXE SRC
☐ Genio 700 510 MT8390 MT8370 Android DRAM CVI CQI Tools	V1.7	2025-03-18	IoT	PDF JLS EXE SRC
☐ Genio 350 MT8365 Yocto DRAM CQI Tools	v1.2	2025-03-18	IoT	PDF JLS EXE SRC
☐ Genio 350 MT8365 Android DRAM CQI Tools	v1.3	2025-03-18	IoT	PDF JLS EXE SRC

Eye Scan: LPDDR4x JEDEC Judgment Method

- TX window
 - DRAM Data Receive mode
- RX window
 - DRAM Data Output mode
- CA window
 - DRAM Command/Address Input Bus

JEDEC Standard No. 209-4B
Page 295

10.6 DQ Rx voltage and timing

JEDEC spec is define at pin

The DQ input receiver mask for voltage and timing is shown Figure 177 is applied per pin. The "total" mask (VdIVW_total, TdIVW_total) defines the area the input signal must not encroach in order for the DQ input receiver to successfully capture an input signal with a BER of lower than tbd. The mask is a receiver property and it is not the valid data-eye.

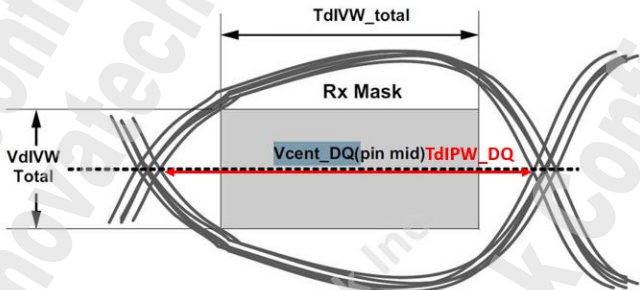


Figure 177 — DQ Receiver(Rx) mask

Symbol	Parameter	1600/1867 ^A		2133/2400		3200		4266		Unit	NOTE
		min	max	min	max	min	max	min	max		
VdIVW_total	Rx Mask voltage - p-p total	-	140	-	140	-	140	-	120	mV	1,2,3,4
TdIVW_total	Rx timing window total (At VdIVW voltage levels)	-	0.22	-	0.22	-	0.25	-	0.25	UI*	1,2,4
TdIVW_1bit	Rx timing window 1 bit toggle (At VdIVW voltage levels)	-	TBD	-	TBD	-	TBD	-	TBD	UI*	1,2,4, 12
VIHL_AC	DQ AC input pulse amplitude pk-pk	180	-	180	-	180	-	170	-	mV	5,13
TdIPW_DQ	Input pulse width (At Vcent_DQ)	0.45	-	0.45	-	0.45	-	0.45	-	UI*	8,6

Eye Scan: LPDDR4x DRAM TX Window in Log Message

- According to JEDEC specifications, TdIPW_DQ ≥ 0.45 UI at the DRAM pad.
 - DRAM should guarantee working properly if the eye is open at pin ≥ 0.45 UI.

```
[TX minimum per bit window]
VrefDQ Range : 0
VrefDQ
[HQALOG] 4266 VrefDQ Channel0 Rank0 30
[HQALOG] 4266 VrefDQ Channel11 Rank0 32
[HQALOG] 4266 VrefDQ Channel12 Rank0 36
[HQALOG] 4266 VrefDQ Channel13 Rank0 32
TX_Window
[HQALOG] 4266 TX_Window Channel0 Rank0 25 (bit 14)
[HQALOG] 4266 TX_Window Channel11 Rank0 26 (bit 1)
[HQALOG] 4266 TX_Window Channel12 Rank0 24 (bit 11)
[HQALOG] 4266 TX_Window Channel13 Rank0 25 (bit 7)
TX_Min_Window(%)
[HQALOG] 4266 TX_Window(%) Channel0 Rank0 79% (PASS)
[HQALOG] 4266 TX_Window(%) Channel11 Rank0 82% (PASS)
[HQALOG] 4266 TX_Window(%) Channel12 Rank0 75% (PASS)
[HQALOG] 4266 TX_Window(%) Channel13 Rank0 79% (PASS)
```

Explanation:

JEDEC Spec : $0.79 \text{ UI} \geq 0.45 \text{ UI}$

Ex: 79% means $0.79 * \text{UI} \geq 0.45 \text{ UI}$

Eye Scan: LPDDR4x DRAM RX Window in log message

- JEDEC proposed RX pass criteria ≥ 0.45 UI

```
[RX minimum per bit window]
Delaycell measurement(/100ps)
[HQALOG] 4266 delaycell 247
VrefRX
[HQALOG] 4266 VrefRX Channel0 Byte0 46
[HQALOG] 4266 VrefRX Channel0 Byte1 44
[HQALOG] 4266 VrefRX Channel1 Byte0 43
[HQALOG] 4266 VrefRX Channel1 Byte1 44
[HQALOG] 4266 VrefRX Channel2 Byte0 38
[HQALOG] 4266 VrefRX Channel2 Byte1 38
[HQALOG] 4266 VrefRX Channel3 Byte0 42
[HQALOG] 4266 VrefRX Channel3 Byte1 44
RX Window
[HQALOG] 4266 RX_Window Channel0 Rank0 76 (bit 7)
[HQALOG] 4266 RX_Window Channel1 Rank0 76 (bit 4)
[HQALOG] 4266 RX_Window Channel2 Rank0 76 (bit 9)
[HQALOG] 4266 RX_Window Channel3 Rank0 76 (bit 1)
RX_Window(%)
[HQALOG] 4266 RX_Window(%) Channel0 Rank0 18772/100ps (81%) (PASS)
[HQALOG] 4266 RX_Window(%) Channel1 Rank0 18772/100ps (81%) (PASS)
[HQALOG] 4266 RX_Window(%) Channel2 Rank0 18772/100ps (81%) (PASS)
[HQALOG] 4266 RX_Window(%) Channel3 Rank0 18772/100ps (81%) (PASS)
```

Explanation:

JEDEC Spec : ≥ 0.45 UI

Ex: 81% means $0.81 \text{ UI} \geq 0.45 \text{ UI}$

4266Mbps: $187.72 \text{ ps} / 234.41\text{ps} \geq 0.45 \text{ UI}$

Eye Scan: LPDDR4x DRAM CA Window in Log Message

- According to JEDEC specifications, TdIPW_DQ = 0.6 UI at the DRAM pad.
 - DRAM must ensure proper operation if the eye is open @ pin ≥ 0.6 UI.

```
[Cmd Bus Training window]
VrefCA
[HQALOG] 4266 VrefCA Channel0 Rank0 22
[HQALOG] 4266 VrefCA Channel1 Rank0 24
[HQALOG] 4266 VrefCA Channel2 Rank0 24
[HQALOG] 4266 VrefCA Channel3 Rank0 24
CA_Window
[HQALOG] 4266 CA_Window Channel0 Rank0 58 (bit 0)
[HQALOG] 4266 CA_Window Channel1 Rank0 58 (bit 5)
[HQALOG] 4266 CA_Window Channel2 Rank0 58 (bit 0)
[HQALOG] 4266 CA_Window Channel3 Rank0 60 (bit 5)
CA_Min_Window(%)
[HQALOG] 4266 CA_Window(%) Channel0 Rank0 91% (PASS)
[HQALOG] 4266 CA_Window(%) Channel1 Rank0 91% (PASS)
[HQALOG] 4266 CA_Window(%) Channel2 Rank0 91% (PASS)
[HQALOG] 4266 CA_Window(%) Channel3 Rank0 94% (PASS)
```

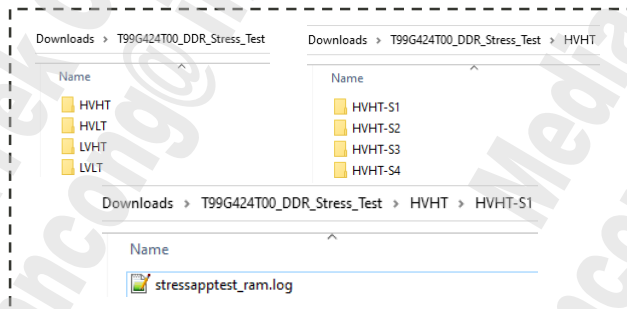
Explanation:

JEDEC Spec : TdIPW (min) ≥ 0.6 UI

Ex: 91% means $0.91 \text{ UI} \geq 0.6 \text{ UI}$

Stress Test (HTLV/LTHV/HTHV/LTLV)

- After passing the eye scan test, start doing the stress test on HT, LT.
- Combination Conditions (HTLV/LTHV/HTHV/LTLV)
 - **At least 4 platforms**
 - Test Images with different (V_{CORE} , V_{MEM})
 - Tool: stressapptest
 - Each combination condition needs to continuously run the stress test for more than **12 hours** without any error message.
- Check Result



```

2021/12/07-20:44:14(GMT) Stats: SAT revision 1.0.7_autoconf, 64 bit binary
2021/12/07-20:44:14(GMT) Log: Android version from open source release
2021/12/07-20:44:14(GMT) Log: 1 nodes, 8 cpus.
2021/12/07-20:44:14(GMT) Log: Prefer plain malloc memory allocation.
2021/12/07-20:44:14(GMT) Log: Using mmap() allocation at 0x75d1c00000.
2021/12/07-20:44:14(GMT) Stats: Starting SAT, 1024M, 5 seconds
2021/12/07-20:44:14(GMT) Log: region number 7 exceeds region count 1
2021/12/07-20:44:14(GMT) Log: Region mask: 0x1
2021/12/07-20:44:19(GMT) Stats: Found 0 hardware incidents
2021/12/07-20:44:19(GMT) Stats: Completed: 116398.00M in 5.00s 23265.74MB/s, with 0 hardware incidents, 0 errors
2021/12/07-20:44:19(GMT) Stats: Memory Copy: 45910.00M at 9178.50MB/s
2021/12/07-20:44:19(GMT) Stats: File Copy: 0.00M at 0.00MB/s
2021/12/07-20:44:19(GMT) Stats: Net Copy: 0.00M at 0.00MB/s
2021/12/07-20:44:19(GMT) Stats: Data Check: 0.00M at 0.00MB/s
2021/12/07-20:44:19(GMT) Stats: Invert Data: 70488.00M at 14098.82MB/s
2021/12/07-20:44:19(GMT) Stats: Disk: 0.00M at 0.00MB/s
2021/12/07-20:44:19(GMT)
2021/12/07-20:44:19(GMT) Status: PASS - please verify no corrected errors
2021/12/07-20:44:19(GMT)

```

Thank you

Questions and Discussions